

Modulating the tumour microenvironment in solid and immunotherapy-resistant tumours

A literature synthesis across 19 mechanistic and interventional axes

261 curated records, 2002-2026 · 74 clinical trials · 21 human translational ·
95 preclinical/mechanistic · 71 reviews · 118 resistance-focused

Overview: what the corpus says about modulating the solid-tumour TME

Scope of the synthesis: 261 curated papers (2002–2026) selected from a 16,224-record Europe PMC harvest, covering 19 mechanistic/interventional axes. 120 papers explicitly concern immunotherapy resistance, immune exclusion or cold-tumour biology.

The organising problem

Checkpoint blockade works where a pre-existing, IFN- γ -driven T-cell response is being restrained by an adaptive brake. The clinical literature settled that framing early — PD-L1 is induced by IFN- γ at sites of active immune attack (PMID 22461641), and response tracks with pre-treatment CD8 infiltration at the invasive margin (PMID 25428505). Every axis in this corpus is therefore an answer to one of three questions:

1. **No T cells were ever primed** → antigen supply, innate sensing, DC licensing (innate_agonists, antigen_presentation_ifn, radiation_chemo_priming, epigenetic_priming).
2. **T cells exist but cannot reach or persist in the tumour** → stroma, vessels, chemokines, metabolism (caf_stroma_ecm, vasculature_hypoxia, trafficking_tls, metabolic_tme).
3. **T cells arrive and are actively switched off** → myeloid and Treg suppression, exhaustion programmes (myeloid_macrophage, treg_suppressive_lymphoid, cytokine_engineering).

clinical_resistance collects the papers that show which of these actually breaks in patients; spatial_profiling collects the methods that let you tell them apart in a given tumour.

Cross-cutting conclusions the corpus supports

- **Resistance is usually TME-level, not antigen-level.** Hard genetic escape does occur — JAK1/2 loss (PMID 27903500), B2M/MHC-I loss (PMID 27433843) — but is a minority of cases. The larger fraction is potentially reversible suppression: myeloid programmes (PMID 39633050), TGF- β -driven exclusion (PMID 29443964), lactate/adenosine metabolism (PMID 36859386, PMID 39137401), and cell-intrinsic exclusion programmes (PMID 30388455).
- **Single-agent TME modulation has failed repeatedly at scale; the failures are informative.** IDO1 inhibition (ECHO-301, PMID 31221619), stromal hyaluronan depletion, and CSF1R monotherapy each validated the target biologically but not clinically. The pattern is that depleting one suppressive node is compensated by another, and that stromal ablation can be actively harmful (PMID 24856586).
- **Localisation, not novelty, is the dominant engineering theme of the last five years.** IL-12, IL-2 and STING agonists were all known to work and all limited by systemic toxicity; the productive work is tumour-restricted delivery — intratumoral injection, regulated gene therapy (PMID 31413142), immunocytokines, TME-responsive nanocarriers, and armoured cell products (PMID 35314843, PMID 38168996).

- **Priming modalities are the most clinically mature TME modulators.** Radiotherapy (PMID 30397353, PMID 34479871), chemotherapy-induced immunogenic cell death, epigenetic priming (PMID 26317466) and neoantigen vaccines (PMID 37165196, PMID 38246194) all have randomised or phase 2 support, and they act on question 1 above rather than trying to out-compete the suppressive network.
- **Microbiome modulation is the clearest example of reversing acquired resistance in humans.** Two independent FMT trials converted anti-PD-1-refractory melanoma to responsive (PMID 33542131, PMID 33303685), now extended to a phase 2 in NSCLC and melanoma (PMID 41606121).
- **Cold tumours segregate by which barrier dominates.** PDAC is a stromal/chemokine exclusion problem (PMID 24277834); glioblastoma is a myeloid-suppression and penetrance problem (PMID 35177622, PMID 32437507); MSS colorectal sits between TGF- β exclusion (PMID 29443964) and multi-agent combination signals (PMID 38871975).
- **Combination is now indexed to measurement.** Spatial and single-cell profiling has moved from descriptive atlases to trajectory-level response prediction (PMID 38194915, PMID 40147443), which is the mechanism by which axis selection can become patient-specific rather than histology-specific.

How to read the rest of the notes

notes/01–notes/13 take each axis group in turn: the mechanism, what has been shown in patients, what failed and why, and the open question. `REPORT.md` is the auto-generated listing of every curated paper by axis and evidence level. `index.tsv` is the machine -readable index, including which records have full text mirrored under `fulltext/`.

Myeloid compartment: TAMs, MDSCs, neutrophils, phagocytosis checkpoints

Mechanism

The myeloid compartment is the numerically dominant immune population in most solid tumours and the most consistently suppressive. The corpus supports three separable functions: (i) recruitment-driven suppression, where CCL2/CSF1 pull monocytes into the tumour and locally polarise them (PMID 21654748, PMID 24056773); (ii) cytokine-mediated T-cell shutdown, most directly shown in glioblastoma where myeloid IL-10 drives intratumoral T-cell dysfunction (PMID 35177622); and (iii) neutrophil plasticity, where TGF- β sets an "N2" suppressive phenotype but early-stage human lung TANs can be stimulatory (PMID 19732719, PMID 25384214) — the compartment is context-dependent, not uniformly hostile.

Human evidence

- CSF1R inhibition remains the cleanest preclinical proof that macrophage re-polarisation, not depletion, drives benefit (glioma, PMID 24056773); clinical translation as monotherapy has been weak.
- IL-8 blockade (HuMax-IL8/BMS-986253) completed phase 1 in metastatic solid tumours and is the most direct human test of the myeloid-recruitment axis (PMID 31488216).
- Prostate cancer provides the strongest recent human resistance story: longitudinal profiling shows myeloid-mediated resistance evolving under therapy rather than being fixed at baseline (PMID 39633050).
- CAR-macrophages have entered the clinic for HER2-overexpressing solid tumours (phase 1, PMID 39920391) — a shift from suppressing myeloid cells to using them as the effector.

What failed and why

Single-node myeloid blockade (CSF1R, CCR2) is compensated: depleting one recruitment axis promotes another, and TAM subsets are spatially and temporally heterogeneous (PMID 39068459). The current framing in the reviews (PMID 37211559, PMID 39780248) is that myeloid targeting is a combination partner, indexed to which subset dominates a given tumour.

Open question

Which myeloid state is the actionable one? Subset-level markers (TREM2, SPP1, MARCO) are better defined than subset-level therapeutics, and no trial in this corpus selects patients by myeloid state.

Suppressive lymphoid compartment: Tregs and T-cell exhaustion

Mechanism

Two distinct problems are often conflated. Tregs are an active suppressive population recruited into tumours — the founding observation being CCR4/CCL22-driven Treg recruitment in ovarian cancer with a survival penalty (PMID 15322536), later generalised to the CD8:Treg ratio as the functional readout (PMID 16344461). Exhaustion is a cell-intrinsic differentiation state of CD8 T cells, driven by chronic antigen and enforced co-inhibitory signalling; the corpus supports co-inhibitory receptor cooperativity rather than redundancy — Tim-3/PD-1 (PMID 20819927) and LAG-3/PD-1 synergising on the same cell to suppress autocrine IFN- γ (PMID 39121848).

Human evidence

- Treg depletion improves vaccine-induced immunity in patients, the earliest clean human proof of the axis (PMID 16308572).
- CTLA-4 blockade shifts the intratumoral effector:Treg ratio in patients (PMID 18818309) — the mechanism now attributed largely to Fc-dependent Treg depletion.
- Anti-TIGIT improves PD-L1 blockade through myeloid and Treg effects rather than direct CD8 de-repression (PMID 38418879), an important reframing given the negative phase 3 TIGIT programmes.
- Fc-engineered CTLA-4 blockade (botensilimab) plus PD-1 shows activity in MSS colorectal cancer, the canonical checkpoint-refractory setting (phase 1, PMID 38871975).
- Neoadjuvant PARPi/chemotherapy in ovarian cancer nominates effector-Treg targeting specifically in homologous-recombination-deficient disease (PMID 38971151).

What failed and why

PD-1 blockade can amplify PD-1⁺ Tregs and drive hyperprogression (PMID 31028147) — the strongest mechanistic warning in this axis that de-repressing an inhibitory receptor is not cell-type selective. Treg-selective targets (CCR8, TNFR2) are attractive for this reason but are represented preclinically here, not clinically.

Open question

Whether exhaustion is best addressed by receptor blockade at all, versus preserving progenitor-exhausted/TCF1⁺ populations upstream — which links this axis to `cell_therapy_tme` (FOXO1 stemness, PMID 38600376) rather than to checkpoint chemistry.

CAFs, ECM and TGF- β -driven immune exclusion

Mechanism

Desmoplastic stroma restricts immunotherapy in two ways: physically, by raising interstitial pressure and collapsing vessels (hyaluronan/collagen), and immunologically, via CAF-derived TGF- β , CXCL12 and antigen-presenting CAF states. The corpus establishes CAF heterogeneity as the central fact: inflammatory (iCAF) versus myofibroblastic (myCAF) populations in PDAC (PMID 28232471), antigen-presenting CAFs (PMID 31197017), and a cross-tissue fibroblast atlas assigning distinct immune-modulatory roles to myofibroblast subtypes (PMID 39303725).

Human evidence

- FAP⁺ CAF-derived CXCL12 excludes T cells from PDAC and its blockade synergises with anti-PD-L1 — the single most cited mechanistic result in this axis (PMID 24277834).
- Spatial/single-cell work in patients links FAP⁺ fibroblasts to SPP1⁺ macrophages as a coordinated exclusion niche in colorectal cancer (PMID 35365629), and identifies a CAF subset restricting T-cell infiltration in HNSCC (PMID 37930937). CAF phenotype associates with outcome in NSCLC (PMID 38242124).
- Bifunctional anti-PD-L1/TGF- β RII trapping is the main clinical instrument here (PMID 32079617); galunisertib plus durvalumab in metastatic PDAC gives the phase 1b read on kinase-level TGF- β inhibition (PMID 33688022).
- Stromal remodelling plus stem-cell-plasticity targeting overcomes chemoresistance in TNBC (PMID 30042390).

What failed and why

Enzymatic stromal ablation improved drug delivery in mice (PEGPH20-type approach, PMID 22439937) but failed clinically, and the mechanistic reason is in this corpus: depleting myofibroblasts/fibrosis outright accelerates PDAC and worsens immune suppression (PMID 24856586). Stroma is partly restraining the tumour; the therapeutic goal is normalisation, not removal.

Open question

Which CAF state to target, with what selectivity. FAP is a marker of multiple states, and no clinical agent in this corpus is subset-selective.

Vasculature, hypoxia and vascular normalisation

Mechanism

Abnormal tumour vasculature is both a barrier to T-cell entry and the source of the hypoxic, acidic milieu that suppresses those T cells that arrive. The corpus supports direct immunological effects of the angiogenic axis rather than purely perfusion effects: VEGF-A upregulates inhibitory checkpoints on intratumoral CD8 T cells (PMID 25601652), and HIF-1 α directly transactivates PD-L1 with MDSC-mediated suppression under hypoxia (PMID 24778419). Hypoxia also confers cytoprotection through the UPR/autophagy axis (PMID 20038797).

Human evidence

- The mechanistic rationale for combining anti-angiogenics with checkpoint blockade is high-endothelial-venule formation and lymphocyte entry (PMID 28404866), reviewed as a general strategy in PMID 29508855.
- HCC is where the combination became standard of care; the corpus captures this through the immunotherapy landscape review (PMID 34764464) and a critical appraisal of guideline-level recommendations (PMID 37535375).
- MSS/pMMR colorectal cancer: randomised phase 2 of anti-PD-1 + HDAC inhibitor + anti-VEGF is a direct test of triple TME modulation in a cold tumour (PMID 38438735).
- HIF-2 α inhibition is now clinically validated in RCC (belzutifan vs everolimus, PMID 39167807) — the only pharmacological hypoxia-pathway success in the corpus.
- Glioblastoma: hypoxic macrophages are nominated as the therapeutic handle for vascular normalisation (PMID 38640932).

What failed and why

CheckMate 143 (nivolumab vs bevacizumab in recurrent GBM) is the negative anchor (PMID 32437507): neither immune de-repression nor anti-angiogenesis alone changes survival in a tumour whose barrier is myeloid and anatomical. Anti-angiogenic dosing is also dose-dependent in the wrong direction — excessive pruning worsens hypoxia, so "normalisation windows" rather than maximal VEGF blockade is the operative concept (PMID 29508855, PMID 36797231).

Open question

There is no clinical assay for the normalisation window; dose and schedule are chosen empirically, which plausibly explains inconsistent immunological benefit across tumour types.

Metabolic TME: adenosine, tryptophan, lactate and lipids

Mechanism

Tumour metabolism suppresses immunity through soluble mediators and through nutrient competition. Four sub-axes appear: the CD39/CD73/A2AR adenosine pathway (PMID 20080644, PMID 36859386); tryptophan catabolism via IDO/TDO (PMID 39472902); lactate and, mechanistically newer, protein/histone lactylation (PMID 39934144, PMID 39883522, PMID 39561764); and lipid/cholesterol handling, where cholesterol modulation potentiates CD8 responses (PMID 26982734) and 25-hydroxycholesterol educates suppressive myeloid cells (PMID 38640930).

Human evidence

- Adenosine: A2AR blockade produced objective responses in treatment-refractory RCC — the clearest clinical signal that a metabolic checkpoint can reverse resistance (PMID 31732494).
- Glycolysis: increased tumour glycolysis characterises resistance to adoptive T-cell therapy in human melanoma/NSCLC samples (PMID 29628419), making metabolic state a candidate selection biomarker for cell therapy.
- Lactylation is emerging as the mechanistic bridge from metabolism to epigenetic immune escape: H3K18 lactylation potentiates immune escape in NSCLC (PMID 39137401, indexed under `clinical_resistance`) and PD-L1 lactylation limits the benefit of a serine/glycine-free diet in colorectal cancer (PMID 39577415).

What failed and why

ECHO-301 — epacadostat plus pembrolizumab in melanoma — is the definitive negative phase 3 in this axis (PMID 31221619). Target engagement was likely incomplete and, more importantly, kynurenine can be produced by multiple enzymes and cell types, so IDO1-selective inhibition does not remove the metabolite. The general lesson the corpus supports: metabolite-level readouts, not enzyme occupancy, should gate these trials.

Open question

Whether dietary/systemic metabolic interventions can be dosed to favour T cells rather than tumour cells; PMID 39577415 shows the same intervention doing both.

Innate sensing agonists and engineered cytokines

Two axes are treated together because they share a single limiting problem: both act on potent, evolutionarily conserved inflammatory circuits whose therapeutic index is set by where the drug acts, not by target validity.

Innate sensing (STING, TLR, RIG-I, type-I IFN)

Mechanistically the corpus is unusually coherent: cytosolic DNA sensing through cGAS-STING is what makes a tumour immunogenic in the first place (PMID 25517615, PMID 27367184, PMID 31852718), micronuclei formed after DNA damage are the physiological source of that signal (PMID 28759889), and therefore DNA-damage-response inhibition promotes antitumour immunity through STING (SCLC, PMID 30777870). Radiotherapy remodels the inflammatory microenvironment through the same route (PMID 32161398).

The clinical problem is delivery and durability: free cyclic dinucleotides are cleared and poorly cell-penetrant, motivating endosomolytic carriers (PMID 30664751), non-CDN STING mimics (PMID 38480836), and metal-ion adjuvants (Mn^{2+} , PMID 32839553). DC-intrinsic STING licensing of cDC1 is where the required activity actually sits (PMID 38363830). Counter-evidence worth keeping: chronic STING activation by mtDNA from senescent tumour cells drives PMN-MDSC suppression (PMID 40203808) — pathway direction depends on duration and cell type, which is the likely reason intratumoral STING agonist trials underdelivered relative to preclinical data.

Engineered cytokines

IL-12 dominates the axis because it is the most potent and the most toxic. The corpus tracks the entire localisation strategy: regulatable IL-12 gene therapy in recurrent high-grade glioma (phase 1, PMID 31413142), immunocytokines for GBM (PMID 33028706), IL-12-armoured CAR-T (PMID 33469002, PMID 37041154), and IL-12 reprogramming of CAR-NKT cells (PMID 38167707). IL-15 superagonist plus nivolumab in PD-1-refractory NSCLC is the main clinical test of a γ c-cytokine as a resistance-reversal agent (ALT-803, PMID 29628312).

The suppressive-cytokine side is equally represented: macrophage IL-10 blocks chemotherapy-dependent CD8 responses by suppressing intratumoral DC IL-12 (PMID 25446896), IL-10 promotes liver metastasis via PD-L1 induction (PMID 38160941), and macrophage IL-10 output is sustained by mitochondrial complex III (PMID 39841829). Note the deliberate tension with `cell_therapy_tme`, where IL-10 expressed by CAR-T cells prevents dysfunction (PMID 38168996): the same cytokine is suppressive in the myeloid compartment and metabolically restorative in the T-cell compartment.

CANTOS provides the outlier result — IL-1 β blockade reducing incident lung cancer in an atherosclerosis population (PMID 28855077) — the only chemoprevention-scale evidence that dampening tumour-promoting inflammation changes cancer incidence.

Open question

For both axes: whether tumour-restricted activation (masked cytokines, prodrugs, regulated vectors, intratumoral routes) can widen the therapeutic index enough for systemic use, and whether pulsatile rather than continuous innate activation avoids the MDSC-inducing chronic phenotype.

T-cell trafficking, chemokines and tertiary lymphoid structures

Mechanism

Infiltration is a chemokine problem with a stromal gatekeeper. The corpus supports the CXCL9/CXCL10-CXCR3 axis as the effector-recruitment route and CXCL12-CXCR4 as the exclusion route (PMID 16269611, PMID 38066335). The decisive mechanistic result is that CAF-derived CXCL12 coats tumour cells and keeps T cells out of PDAC (PMID 24277834); conversely, CXCL9/10-engineered DCs increase infiltration and checkpoint-blockade efficacy in lung cancer (PMID 38518770).

Tertiary lymphoid structures (TLS) reframe the problem: rather than importing effectors, generate them locally. TLS and B-cell content predict immunotherapy response across melanoma, RCC and sarcoma (PMID 31942075, PMID 31942077), with T follicular helper infiltration as an earlier prognostic hint (PMID 23778140). Induction is now achievable preclinically — IL-33-activated ILC2s induce TLS in pancreatic cancer (PMID 39814891) — and clinically approachable via Fc-optimised CD40 agonism, which elicits TLS formation and systemic antitumour immunity in patients (PMID 40816292).

Human evidence

CD8 infiltration is an independent favourable prognostic indicator (basal-like breast, PMID 22420471), and the infiltration/PD-L1 cross-classification (PMID 25977340) is the scheme that still organises trial stratification into hot/altered/cold tumours. CXCR4 antagonism plus pembrolizumab and chemotherapy in PDAC (COMBAT/BL-8040, PMID 32451495) is the main clinical test of blocking the exclusion route.

What failed and why

CXCR4 blockade produced modest single-arm activity in PDAC and has not translated into randomised benefit; the corpus suggests why — chemokine gradients are redundant, and removing one exclusion signal does not create the CXCL9/10 gradient needed to import effectors. The productive framing is that trafficking interventions must be paired with a priming source (see notes/08, notes/09).

Open question

TLS induction has no accepted pharmacological standard, no agreed maturity readout, and unclear durability after the inducing agent stops.

Antigen presentation, MHC-I loss, IFN- γ pathway defects and neoantigen therapy

Mechanism

This axis contains the least reversible resistance in the corpus, because the defects are in the tumour cell's ability to be seen at all. Three classes appear:

1. **IFN- γ sensing loss** — JAK1/2 loss-of-function causes primary resistance to PD-1 blockade (PMID 27903500).
2. **MHC-I loss** — genetic (B2M) and, importantly, non-genetic: PDAC degrades MHC-I through NBR1-dependent autophagy, which is pharmacologically restorable (PMID 32376951). Reviewed as a target class in PMID 38041084.
3. **Oncogenic-pathway-driven exclusion** — STK11/LKB1 deficiency drives neutrophil recruitment and suppressive cytokines (PMID 26833127) and predicts PD-1 inhibitor resistance in KRAS-mutant lung adenocarcinoma (PMID 29773717), the most actionable genotype-level resistance marker in routine practice.

Radiation is the counter-mechanism with the longest history: it broadens the presented peptide repertoire and raises MHC-I (PMID 16636135), linking this axis to notes/09.

Human evidence: neoantigen therapy

If antigen visibility is the problem, supplying antigen is the intervention. The corpus tracks a clean progression: first-in-human personalised neoantigen vaccine in melanoma (PMID 28678778); individualised mRNA-4157/V940 plus pembrolizumab improving recurrence-free survival versus pembrolizumab in resected melanoma (randomised phase 2, PMID 38246194); autogene-gevemeran-type RNA neoantigen vaccines priming T cells in resected PDAC (PMID 37165196) with follow-up showing those CD8 clones are long-lived (PMID 39972124); and a neoantigen vaccine generating immunity in RCC — a low-TMB tumour (PMID 39910301).

What failed and why

Nothing in this axis has a definitive negative phase 3 in the corpus, which is itself the point: the field is early and the evidence is favourable but small-N. The structural risk is that vaccination presumes intact presentation and IFN- γ signalling — precisely what is broken in the resistance papers above. Patients selected for vaccine trials by TMB rather than by presentation competence may be the wrong denominator.

Open question

Whether MHC-I restoration (autophagy inhibition, epigenetic de-repression) should be a prerequisite arm for neoantigen therapy in PDAC-like tumours.

Priming the cold tumour: epigenetic, radiation and chemotherapy approaches

Epigenetic priming

The mechanism is viral mimicry: DNA methyltransferase inhibition de-represses endogenous retroviruses, generating dsRNA and a cell-intrinsic interferon response (PMID 26317466). The corollary is that tumours actively silence retroelements to stay invisible — KDM5B recruits SETDB1 to do exactly that, and its loss restores immunogenicity (PMID 34671158); HIF-regulated ERV translation adds a hypoxia-linked route in RCC (PMID 40023154). LSD1 ablation enables checkpoint blockade through the same dsRNA-stress logic (PMID 29937226). HDAC inhibition is a separate, less clean mechanism — it raises PD-1 ligands and augments PD-1 blockade (PMID 26297712).

Human evidence is phase 2-scale: epigenetic potentiation of NY-ESO-1 vaccination in ovarian cancer (PMID 24535937); pembrolizumab plus entinostat in metastatic uveal melanoma (PEMDAC, PMID 34453044); entinostat plus nivolumab in metastatic PDAC (PMID 39532835). TNBC is the tumour type where the axis is most developed conceptually (PMID 32461152, PMID 38719195, PMID 40155623), and EZH1/EZH2 inhibition is being used to improve adoptive T cells rather than the tumour (PMID 39983725).

Why results are modest: class-wide epigenetic drugs act on T cells as well as tumour cells, and the therapeutic window between de-repressing ERVs and impairing effector differentiation is narrow.

Radiation priming

Radiotherapy is the most clinically mature TME modulator here. Fractionated — not single-dose — radiotherapy plus CTLA-4 blockade produces abscopal responses (PMID 19706802), and this translated: radiotherapy induces responses to CTLA-4 blockade in chemorefractory NSCLC (PMID 30397353), SBRT plus pembrolizumab improves response over pembrolizumab alone in metastatic NSCLC (randomised phase 2, PMID 31294749), and pembrolizumab plus radiotherapy induces systemic responses in immunologically cold NSCLC (PMID 40696153). Low-dose radiotherapy reverses immune desertification and resistance — a mechanistically distinct, infiltration-oriented use of radiation (PMID 34479871).

Negative anchors matter: ipilimumab after radiotherapy in mCRPC missed its primary endpoint (PMID 24831977), and adding nivolumab to chemoradiotherapy in MGMT-unmethylated newly diagnosed glioblastoma failed in phase 3 (CheckMate 498-type design, PMID 35511454). Radiation also creates suppression — a post-radiotherapy TME with recurrence-promoting features (PMID 26105538), so schedule and dose determine whether radiation primes or immunosuppresses.

Chemotherapy and immunogenic cell death (ICD)

ICD is the unifying concept: dying tumour cells release calreticulin/ATP/HMGB1 and provide both antigen and adjuvant. The corpus covers nanoparticle-mediated ICD (PMID 30016571,

PMID 29675465), necroptosis-directed ICD (PMID 38454135) and PDT-induced ICD (PMID 31842994). Human mechanistic depth comes from single-cell/spatial profiling of pembrolizumab plus radiation in TNBC, which resolves three distinct response trajectories rather than a binary outcome (PMID 38194915).

Open question

Dose/fraction/timing optimisation is unsolved and is the main reason radio-immunotherapy trials disagree; the corpus contains no head-to-head schedule comparison in humans.

Local biological modulation: intratumoral therapy, oncolytic viruses, engineered bacteria, microbiome

Intratumoral and oncolytic approaches

The rationale is the same localisation argument as for cytokines, executed with a replicating agent. GM-CSF-armed oncolytic adenovirus established the template (PMID 20484030); oncolytic virotherapy increases intratumoral T-cell infiltration and improves anti-PD-1 activity (PMID 28886381), which is the mechanistic claim that justifies the combination. Engineered bacteria are the parallel platform (Salmonella secreting heterologous flagellin, PMID 28179508).

Glioblastoma is where the clinical evidence concentrates, because local delivery is already the surgical norm: G207 in paediatric high-grade glioma (PMID 33838625), a CEA-expressing measles derivative in recurrent GBM (phase 1, PMID 38216554), and a trial linking oncolytic immunoactivation to survival (PMID 37853118). Neoadjuvant anti-PD-1 in GBM shows intratumoral plus systemic immune responses with a survival signal (PMID 30742122) — evidence that timing, not just route, is a modulation variable. VG161 has activity in refractory HCC (PMID 40108464).

Mechanistically the most interesting recent result is that elimination of solid tumours requires intratumoral immune triads — CD4/CD8/DC co-localisation (PMID 38906155) — which explains why importing a single cell type or agonist is insufficient. The counterweight is that the TME represses T-cell mitochondrial biogenesis, causing intratumoral metabolic insufficiency (PMID 27496732): a locally injected agent still delivers effectors into a hostile metabolic environment.

Microbiome modulation

This is the corpus's clearest demonstration that acquired immunotherapy resistance can be reversed in humans by modifying the host rather than the tumour. Preclinical anchor: the efficacy of CTLA-4 blockade depends on gut microbiota (PMID 26541610). Two independent FMT trials converted anti-PD-1-refractory melanoma patients to responders (PMID 33542131, PMID 33303685); a subsequent phase 1 confirmed feasibility (PMID 37414899) and a phase 2 (FMT-LUMINate) extends the approach to NSCLC and melanoma (PMID 41606121). Gut bacterial composition also drives primary resistance in RCC (PMID 32376136, indexed under `clinical_resistance`).

The intratumoral microbiome is a distinct problem: pancreatic tumour bacteria induce innate and adaptive suppression (PMID 29567829) and tumour microbiome diversity tracks PDAC outcomes (PMID 31398337) — with the therapeutic twist that killing intratumoral bacteria with a liposomal antibiotic generates immunogenic neoantigens (PMID 37749267).

Open question

Donor selection, engraftment durability and mechanism attribution for FMT remain unresolved; there is no defined bacterial consortium in this corpus with clinical equivalence to whole-stool transplant.

TME-directed delivery, regulated cell death, and physical/energy-based modulation

TME-responsive delivery and regulated cell death

Two ideas are combined in this axis. The first is stimulus-responsive delivery: carriers that decompose in the tumour's acidic, H_2O_2 -rich, hypoxic milieu, exemplified by hollow MnO_2 platforms that relieve hypoxia while delivering payload (PMID 29026068), and reviewed as precision nanomedicine for TME navigation (PMID 40457437).

The second is which death pathway to induce, because the immunological consequences differ. Ferroptosis is bidirectional in this corpus and that is its defining feature: CD8 T cells kill in part by inducing tumour ferroptosis (PMID 31043744), yet ferroptosis of tumour-infiltrating neutrophils causes immune suppression (PMID 36385526). Therapeutic exploitation is therefore selectivity-dependent — tumour-specific GPX4 degradation in PDAC (PMID 37910602), CPT1A inhibition synergising with immunotherapy in lung cancer (PMID 38453925), ferroptosis heterogeneity guiding combinations in TNBC (PMID 36257316), and a metastatic gastric cancer atlas linking ferroptosis to immunotherapy response (PMID 39097198). Pyroptosis is the parallel route: mitochondria-targeted PDT triggering GSDME-mediated pyroptosis sensitises anti-PD-1 in colorectal cancer (PMID 38429070), and PARP inhibitors trigger TNF-caspase-8-GSDMD/E pyroptosis in ovarian cancer (PMID 39366751) — a mechanism relevant to already-approved drugs.

Physical and energy-based modulation

Energy deposition is a way to convert a cold tumour locally without systemic exposure. Hyperthermia is the historical precedent (PMID 12147435); cryoablation plus CTLA-4 blockade is the cleanest preclinical proof of in-situ vaccination by physical means (PMID 22108823). Photodynamic and photothermal therapy are the largest sub-literature (PMID 33431631, PMID 37129253), including nanoscale coordination polymers combining chemotherapy and PDT to potentiate checkpoint blockade (PMID 27530650); NIR photoimmunotherapy has reached clinical use in HNSCC (PMID 31335117). Sonodynamic therapy extends the same logic to depth (PMID 32337527, PMID 37914681, PMID 39735354), and focused ultrasound has been used clinically for transcranial BBB opening and immunostimulation in brain tumours (PMID 33547073, with a GTP-activatable immunosonodynamic probe in PMID 39613729).

Limits

Both axes are dominated by preclinical work with device- and formulation-specific results, and depth/access constrain clinical use (light penetration for PDT, acoustic window for SDT). The corpus contains no randomised evidence that adding energy-based modulation to checkpoint blockade improves survival.

Open question

Whether a physically primed single lesion produces reliable systemic (abscopal) immunity — the same unresolved question as radiation priming in notes/09, with less clinical data.

Cell therapy against the solid-tumour TME

Why this belongs in a TME corpus

Adoptive cell therapy fails in solid tumours largely for TME reasons rather than targeting reasons, so cell-therapy engineering has become a form of TME modulation carried in the effector cell.

What patients have shown

- HER2 CAR-T in sarcoma: persistence without adequate expansion or tumour control — the founding solid-tumour limitation (PMID 25800760).
- EGFRvIII CAR-T in glioblastoma: a single infusion drove antigen loss and induced adaptive resistance (Treg and inhibitory-molecule upregulation) in resected tumour — the single most instructive human cell-therapy TME result (PMID 28724573). Repeated infusion plus pembrolizumab showed no efficacy (PMID 38216766), confirming that neither redosing nor checkpoint blockade fixes it.
- Armouring works in humans: PSMA-targeting, TGF- β -insensitive CAR-T produced responses in mCRPC (phase 1, PMID 35314843) — direct evidence that neutralising one dominant TME signal changes clinical activity.
- TIL therapy: lifileucel monotherapy has activity in advanced NSCLC, including ICI-refractory disease (PMID 38563600).
- CAR-macrophages have entered the clinic for HER2⁺ solid tumours (PMID 39920391), building on human CAR-M preclinical work (PMID 32361713).

Engineering strategies represented

Signalling rewiring (JAK-STAT-containing CAR, PMID 29400710), autocrine immunometabolic support (IL-10-expressing CAR-T resisting dysfunction and clearing metastatic solid tumours, PMID 38168996), transcriptional stemness enforcement (FOXO1, PMID 38600376), and senolytic CAR-T targeting uPAR⁺ senescent cells — a non-tumour-cell target within the microenvironment (PMID 32555459).

Limits

Reviews converge on the same list: antigen heterogeneity and loss, poor trafficking, exhaustion, and metabolic insufficiency in the TME (PMID 33824268, PMID 36717905, PMID 40610404). Nothing here is a solved problem in solid tumours; the phase 1 signals above are early and small.

Open question

Whether cell-intrinsic armouring (TGF- β insensitivity, IL-10, FOXO1) can substitute for modifying the host TME pharmacologically, or whether both are required — the corpus currently supports "both" only by inference, not by comparative trial data.

Immunotherapy resistance: mechanisms in patients, and how the TME is measured

The mechanistic taxonomy the corpus supports

Adaptive (IFN- γ -induced) resistance. PD-L1 is expressed at sites of active immune attack (PMID 12091876, PMID 22461641) and PD-1 blockade works by releasing that brake (PMID 25428505). This is the resistance that immunotherapy already treats.

Oncogene-driven immune evasion. PTEN loss increases PD-L1 and confers immunoresistance in glioma (PMID 17159987) and promotes resistance to T-cell-mediated therapy in melanoma (PMID 26645196); EGFR activation engages the PD-1 axis in lung tumours (PMID 24078774); β -catenin activation drives anti-PD-1 resistance in HCC (PMID 31186238); ZEB1 promotes immune escape in melanoma (PMID 35288462); TGF- β drives evasion in reconstituted colorectal metastasis (PMID 29443964).

Hard genetic escape. Acquired-resistance mutations in melanoma including B2M loss (PMID 27433843) and JAK1/2 loss (see notes/08) — the subset least likely to be rescued by TME modulation.

Metabolic/epigenetic escape. H3K18 lactylation potentiates immune escape in NSCLC (PMID 39137401) with ACSS2 identified as a lactyl-CoA synthetase coupling KAT2A to histone lactylation (PMID 39561764) — a mechanism that ties nutrient state directly to chromatin-level immune evasion. Mitochondrial transfer from tumour to T cells is a newer route (PMID 39843734).

Host and organ context. Gut bacterial composition drives primary resistance in RCC (PMID 32376136); NASH-associated liver biology limits anti-tumour surveillance in immunotherapy-treated HCC (PMID 33762733) — resistance determined outside the tumour altogether.

Programme-level exclusion. T-cell exclusion is encoded as a cancer-cell programme (PMID 30388455) and as an inferable signature of dysfunction/exclusion predicting response (PMID 30127393); a pan-cancer T-cell atlas links a stress-response state to resistance (PMID 37248301). The therapeutic framing — hot, altered, cold — comes from PMID 30610226, and low-dose radiotherapy reversing immune desertification (PMID 34479871) is the corpus's proof that "cold" is a modifiable state.

Measurement: what makes axis selection possible

Spatial and single-cell profiling moved from atlases (breast, PMID 30982598; GBM progression, PMID 35624211) to therapy-linked inference: spatial multi-omics of immunotherapy efficacy in advanced NSCLC (PMID 36854570), ecosystem dynamics under neoadjuvant chemo-immunotherapy in oesophageal cancer (PMID 36921563), an anti-PD-1-treated NSCLC single-cell atlas (PMID 40147443), pan-cancer stromal target discovery (PMID 38225927), conserved CAF spatial subtypes (PMID 40154487), and TIM-3 nominated from lung adenocarcinoma precursors (PMID 40345189). Multi-omic ML predictors of therapy response exist (PMID 34875674) but are single-cohort.

The practical conclusion

Resistance mechanisms are heterogeneous within histology, and most of the reversible ones are TME-level. That argues for assigning a TME axis per patient — myeloid-high, stroma-excluded, metabolically suppressed, presentation-deficient — using the profiling tools above, rather than combining agents by tumour type. No trial in this corpus allocates therapy that way; that gap is the most consistent unmet need across all 19 axes.

TME modulation in solid and immunotherapy-resistant tumours — evidence listing

261 curated records, 2002–2026. 118 (45%) explicitly address immunotherapy resistance, immune exclusion or cold-tumour biology; 112 have open-access full text mirrored under [fulltext/](#) (marked FT).

Evidence mix — Clinical trials: 74, Human translational: 21, Evidence synthesis: 1, Reviews: 70, Preclinical (in vivo): 75, Mechanistic: 20

Disease context — pan-solid: 88, melanoma: 45, nsclc: 27, gbm_cns: 25, breast: 22, pdac: 17, crc: 16, gu_other: 13, gyn: 8, rcc: 7, sarcoma_peds: 7, hcc_biliary: 6, gi_other: 5, hnscc: 2, mesothelioma: 1

Cross-cutting TME modulation & immune contexture

22 papers · 2010–2025 · 5 resistance-focused

Clinical trials

- **2010** Kantoff PW et al. — Sipuleucel-T immunotherapy for castration-resistant prostate cancer. The New England journal of medicine. [PMID 20818862](#)
- **2014** Tumei PC et al. — PD-1 blockade induces responses by inhibiting adaptive immune resistance. Nature. [PMID 25428505](#) FT
- **2014** Powles T et al. — MPDL3280A (anti-PD-L1) treatment leads to clinical activity in metastatic bladder cancer. Nature. [PMID 25428503](#)
- **2020** Petitprez F et al. — B cells are associated with survival and immunotherapy response in sarcoma. Nature. [PMID 31942077](#)
- **2021** Pfister D et al. — NASH limits anti-tumour surveillance in immunotherapy-treated HCC. Nature. [PMID 33762733](#) FT
- **2022** Cercek A et al. — PD-1 Blockade in Mismatch Repair-Deficient, Locally Advanced Rectal Cancer. The New England journal of medicine. [PMID 35660797](#)
- **2025** Reiss KA et al. — CAR-macrophage therapy for HER2-overexpressing advanced solid tumors: a phase 1 trial. Nature medicine. [PMID 39920391](#)

Human translational

- **2016** Peng W et al. — Loss of PTEN Promotes Resistance to T Cell-Mediated Immunotherapy. Cancer discovery. [PMID 26645196](#)
- **2018** Jiang P et al. — Signatures of T cell dysfunction and exclusion predict cancer immunotherapy response. Nature medicine. [PMID 30127393](#)
- **2018** Jerby-Arnon L et al. — A Cancer Cell Program Promotes T Cell Exclusion and Resistance to Checkpoint Blockade. Cell. [PMID 30388455](#)

Reviews

- **2010** Kessenbrock K et al. — Matrix metalloproteinases: regulators of the tumor microenvironment. *Cell*. [PMID 20371345](#)
- **2011** Mellman I et al. — Cancer immunotherapy comes of age. *Nature*. [PMID 22193102](#)
- **2012** Hanahan D et al. — Accessories to the crime: functions of cells recruited to the tumor microenvironment. *Cancer cell*. [PMID 22439926](#)
- **2013** Gajewski TF et al. — Innate and adaptive immune cells in the tumor microenvironment. *Nature immunology*. [PMID 24048123](#)
- **2018** Binnewies M et al. — Understanding the tumor immune microenvironment (TIME) for effective therapy. *Nature medicine*. [PMID 29686425](#)
- **2019** Hinshaw DC et al. — The Tumor Microenvironment Innately Modulates Cancer Progression. *Cancer research*. [PMID 31350295](#)
- **2023** de Visser KE et al. — The evolving tumor microenvironment: From cancer initiation to metastatic outgrowth. *Cancer cell*. [PMID 36917948](#)
- **2024** Wu B et al. — Cold and hot tumors: from molecular mechanisms to targeted therapy. *Signal transduction and targeted therapy*. [PMID 39420203](#) FT
- **2025** Zhang M et al. — Advances in cancer immunotherapy: historical perspectives, current developments, and future directions. *Molecular cancer*. [PMID 40336045](#) FT

Preclinical (in vivo)

- **2014** Özdemir BC et al. — Depletion of carcinoma-associated fibroblasts and fibrosis induces immunosuppression and accelerates pancreas cancer with reduced survival. *Cancer cell*. [PMID 24856586](#)
- **2015** Chang CH et al. — Metabolic Competition in the Tumor Microenvironment Is a Driver of Cancer Progression. *Cell*. [PMID 26321679](#)
- **2018** Böttcher JP et al. — NK Cells Stimulate Recruitment of cDC1 into the Tumor Microenvironment Promoting Cancer Immune Control. *Cell*. [PMID 29429633](#) FT

Mechanisms of immunotherapy resistance (primary & acquired)

18 papers · 2002-2025 · 18 resistance-focused

Clinical trials

- **2020** Derosa L et al. — Gut Bacteria Composition Drives Primary Resistance to Cancer Immunotherapy in Renal Cell Carcinoma Patients. *European urology*. [PMID 32376136](#)
- **2022** Herrera FG et al. — Low-Dose Radiotherapy Reverses Tumor Immune Desertification and Resistance to Immunotherapy. *Cancer discovery*. [PMID 34479871](#) FT
- **2022** Plaschka M et al. — ZEB1 transcription factor promotes immune escape in melanoma. *Journal for immunotherapy of cancer*. [PMID 35288462](#) FT

Human translational

- **2016** Chen PL et al. — Analysis of Immune Signatures in Longitudinal Tumor Samples Yields Insight into Biomarkers of Response and Mechanisms of Resistance to Immune Checkpoint Blockade. *Cancer discovery*. [PMID 27301722](#)

- **2023** Chu Y et al. — Pan-cancer T cell atlas links a cellular stress response state to immunotherapy resistance. *Nature medicine*. [PMID 37248301](#)

Reviews

- **2019** Jiang X et al. — Role of the tumor microenvironment in PD-L1/PD-1-mediated tumor immune escape. *Molecular cancer*. [PMID 30646912](#) FT
- **2019** Galon J et al. — Approaches to treat immune hot, altered and cold tumours with combination immunotherapies. *Nature reviews. Drug discovery*. [PMID 30610226](#)
- **2025** Tufail M et al. — Immune evasion in cancer: mechanisms and cutting-edge therapeutic approaches. *Signal transduction and targeted therapy*. [PMID 40739089](#) FT

Preclinical (in vivo)

- **2002** Dong H et al. — Tumor-associated B7-H1 promotes T-cell apoptosis: a potential mechanism of immune evasion. *Nature medicine*. [PMID 12091876](#)
- **2013** Akbay EA et al. — Activation of the PD-1 pathway contributes to immune escape in EGFR-driven lung tumors. *Cancer discovery*. [PMID 24078774](#)
- **2018** Tauriello DVF et al. — TGF β drives immune evasion in genetically reconstituted colon cancer metastasis. *Nature*. [PMID 29443964](#)
- **2019** Ruiz de Galarreta M et al. — β -Catenin Activation Promotes Immune Escape and Resistance to Anti-PD-1 Therapy in Hepatocellular Carcinoma. *Cancer discovery*. [PMID 31186238](#)
- **2024** Zhang C et al. — H3K18 Lactylation Potentiates Immune Escape of Non-Small Cell Lung Cancer. *Cancer research*. [PMID 39137401](#)
- **2025** Ikeda H et al. — Immune evasion through mitochondrial transfer in the tumour microenvironment. *Nature*. [PMID 39843734](#) FT

Mechanistic

- **2007** Parsa AT et al. — Loss of tumor suppressor PTEN function increases B7-H1 expression and immunoresistance in glioma. *Nature medicine*. [PMID 17159987](#)
- **2012** Taube JM et al. — Colocalization of inflammatory response with B7-h1 expression in human melanocytic lesions supports an adaptive resistance mechanism of immune escape. *Science translational medicine*. [PMID 22461641](#)
- **2016** Zaretsky JM et al. — Mutations Associated with Acquired Resistance to PD-1 Blockade in Melanoma. *The New England journal of medicine*. [PMID 27433843](#)
- **2025** Zhu R et al. — ACSS2 acts as a lactyl-CoA synthetase and couples KAT2A to function as a lactyltransferase for histone lactylation and tumor immune evasion. *Cell metabolism*. [PMID 39561764](#)

Myeloid compartment: TAMs, MDSCs, neutrophils, phagocytosis checkpoints

13 papers · 2009–2025 · 6 resistance-focused

Clinical trials

- **2014** Eruslanov EB et al. — Tumor-associated neutrophils stimulate T cell responses in early-stage human lung cancer. The Journal of clinical investigation. [PMID 25384214](#)
- **2019** Bilusic M et al. — Phase I trial of HuMax-IL8 (BMS-986253), an anti-IL-8 monoclonal antibody, in patients with metastatic or unresectable solid tumors. Journal for immunotherapy of cancer. [PMID 31488216](#) FT
- **2022** Ravi VM et al. — T-cell dysfunction in the glioblastoma microenvironment is mediated by myeloid cells releasing interleukin-10. Nature communications. [PMID 35177622](#) FT

Human translational

- **2025** Lyu A et al. — Evolution of myeloid-mediated immunotherapy resistance in prostate cancer. Nature. [PMID 39633050](#) FT

Reviews

- **2012** Gabrilovich DI et al. — Coordinated regulation of myeloid cells by tumours. Nature reviews. Immunology. [PMID 22437938](#)
- **2023** Chen S et al. — Macrophages in immunoregulation and therapeutics. Signal transduction and targeted therapy. [PMID 37211559](#) FT
- **2024** Kundu M et al. — Modulation of the tumor microenvironment and mechanism of immunotherapy-based drug resistance in breast cancer. Molecular cancer. [PMID 38715072](#) FT
- **2024** Chu X et al. — Decoding the spatiotemporal heterogeneity of tumor-associated macrophages. Molecular cancer. [PMID 39068459](#) FT
- **2025** He S et al. — Myeloid-derived suppressor cells (MDSCs) in the tumor microenvironment and their targeting in cancer therapy. Molecular cancer. [PMID 39780248](#) FT

Preclinical (in vivo)

- **2011** Qian BZ et al. — CCL2 recruits inflammatory monocytes to facilitate breast-tumour metastasis. Nature. [PMID 21654748](#)
- **2013** Pyonteck SM et al. — CSF-1R inhibition alters macrophage polarization and blocks glioma progression. Nature medicine. [PMID 24056773](#) FT
- **2014** Noy R et al. — Tumor-associated macrophages: from mechanisms to therapy. Immunity. [PMID 25035953](#)

Mechanistic

- **2009** Fridlender ZG et al. — Polarization of tumor-associated neutrophil phenotype by TGF-beta: "N1" versus "N2" TAN. Cancer cell. [PMID 19732719](#)

Suppressive lymphoid compartment: Tregs & T-cell exhaustion

13 papers · 2004–2024 · 3 resistance-focused

Clinical trials

- **2005** Dannull J et al. — Enhancement of vaccine-mediated antitumor immunity in cancer patients after depletion of regulatory T cells. *The Journal of clinical investigation*. PMID [16308572](#)
- **2008** Liakou CI et al. — CTLA-4 blockade increases IFN γ -producing CD4+ICOS^{hi} cells to shift the ratio of effector to regulatory T cells in cancer patients. *Proceedings of the National Academy of Sciences of the United States of America*. PMID [18818309](#)
- **2024** Bullock AJ et al. — Botensilimab plus balstilimab in relapsed/refractory microsatellite stable metastatic colorectal cancer: a phase 1 trial. *Nature medicine*. PMID [38871975](#) FT
- **2024** Guan X et al. — Anti-TIGIT antibody improves PD-L1 blockade through myeloid and Treg cells. *Nature*. PMID [38418879](#) FT
- **2024** Luo Y et al. — Neoadjuvant PARPi or chemotherapy in ovarian cancer informs targeting effector Treg cells for homologous-recombination-deficient tumors. *Cell*. PMID [38971151](#)

Reviews

- **2006** Zou W — Regulatory T cells, tumour immunity and immunotherapy. *Nature reviews. Immunology*. PMID [16557261](#)
- **2020** Li C et al. — Regulatory T cells in tumor microenvironment: new mechanisms, potential therapeutic strategies and future prospects. *Molecular cancer*. PMID [32680511](#) FT
- **2021** Nishikawa H et al. — Mechanisms of regulatory T cell infiltration in tumors: implications for innovative immune precision therapies. *Journal for immunotherapy of cancer*. PMID [34330764](#) FT

Preclinical (in vivo)

- **2004** Curiel TJ et al. — Specific recruitment of regulatory T cells in ovarian carcinoma fosters immune privilege and predicts reduced survival. *Nature medicine*. PMID [15322536](#)
- **2010** Sakuishi K et al. — Targeting Tim-3 and PD-1 pathways to reverse T cell exhaustion and restore anti-tumor immunity. *The Journal of experimental medicine*. PMID [20819927](#) FT
- **2019** Kamada T et al. — PD-1+ regulatory T cells amplified by PD-1 blockade promote hyperprogression of cancer. *Proceedings of the National Academy of Sciences of the United States of America*. PMID [31028147](#) FT
- **2024** Andrews LP et al. — LAG-3 and PD-1 synergize on CD8+ T cells to drive T cell exhaustion and hinder autocrine IFN- γ -dependent anti-tumor immunity. *Cell*. PMID [39121848](#)

Mechanistic

- **2005** Sato E et al. — Intraepithelial CD8+ tumor-infiltrating lymphocytes and a high CD8+/regulatory T cell ratio are associated with favorable prognosis in ovarian cancer. *Proceedings of the National Academy of Sciences of the United States of America*. PMID [16344461](#)

CAFs, ECM & TGF- β -driven exclusion

13 papers · 2012–2024 · 5 resistance-focused

Clinical trials

- **2018** Cazet AS et al. — Targeting stromal remodeling and cancer stem cell plasticity overcomes chemoresistance in triple negative breast cancer. Nature communications. PMID 30042390 FT
- **2020** Lind H et al. — Dual targeting of TGF- β and PD-L1 via a bifunctional anti-PD-L1/TGF- β R2 agent: status of preclinical and clinical advances. Journal for immunotherapy of cancer. PMID 32079617 FT
- **2021** Melisi D et al. — Safety and activity of the TGF β receptor I kinase inhibitor galunisertib plus the anti-PD-L1 antibody durvalumab in metastatic pancreatic cancer. Journal for immunotherapy of cancer. PMID 33688022 FT

Human translational

- **2022** Qi J et al. — Single-cell and spatial analysis reveal interaction of FAP+ fibroblasts and SPP1+ macrophages in colorectal cancer. Nature communications. PMID 35365629 FT
- **2024** Cords L et al. — Cancer-associated fibroblast phenotypes are associated with patient outcome in non-small cell lung cancer. Cancer cell. PMID 38242124 FT
- **2024** Li C et al. — Spatial and Single-Cell Transcriptomics Reveal a Cancer-Associated Fibroblast Subset in HNSCC That Restricts Infiltration and Antitumor Activity of CD8+ T Cells. Cancer research. PMID 37930937 FT

Reviews

- **2016** Kalluri R — The biology and function of fibroblasts in cancer. Nature reviews. Cancer. PMID 27550820
- **2020** Sahai E et al. — A framework for advancing our understanding of cancer-associated fibroblasts. Nature reviews. Cancer. PMID 31980749 FT
- **2021** Mao X et al. — Crosstalk between cancer-associated fibroblasts and immune cells in the tumor microenvironment: new findings and future perspectives. Molecular cancer. PMID 34635121 FT

Preclinical (in vivo)

- **2012** Provenzano PP et al. — Enzymatic targeting of the stroma ablates physical barriers to treatment of pancreatic ductal adenocarcinoma. Cancer cell. PMID 22439937
- **2017** Öhlund D et al. — Distinct populations of inflammatory fibroblasts and myofibroblasts in pancreatic cancer. The Journal of experimental medicine. PMID 28232471 FT
- **2019** Elyada E et al. — Cross-Species Single-Cell Analysis of Pancreatic Ductal Adenocarcinoma Reveals Antigen-Presenting Cancer-Associated Fibroblasts. Cancer discovery. PMID 31197017

Mechanistic

- **2024** Gao Y et al. — Cross-tissue human fibroblast atlas reveals myofibroblast subtypes with distinct roles in immune modulation. *Cancer cell*. [PMID 39303725](#)

Vasculature, hypoxia & vascular normalisation

13 papers · 2010–2024 · 6 resistance-focused

Clinical trials

- **2020** Reardon DA et al. — Effect of Nivolumab vs Bevacizumab in Patients With Recurrent Glioblastoma: The CheckMate 143 Phase 3 Randomized Clinical Trial. *JAMA oncology*. [PMID 32437507](#) FT
- **2022** Llovet JM et al. — Immunotherapies for hepatocellular carcinoma. *Nature reviews. Clinical oncology*. [PMID 34764464](#)
- **2024** Wang F et al. — Combined anti-PD-1, HDAC inhibitor and anti-VEGF for MSS/pMMR colorectal cancer: a randomized phase 2 trial. *Nature medicine*. [PMID 38438735](#)
- **2024** Choueiri TK et al. — Belzutifan versus Everolimus for Advanced Renal-Cell Carcinoma. *The New England journal of medicine*. [PMID 39167807](#)

Human translational

- **2024** Wang W et al. — Identification of hypoxic macrophages in glioblastoma with therapeutic potential for vasculature normalization. *Cancer cell*. [PMID 38640932](#)

Evidence synthesis

- **2024** Cappuyns S et al. — Critical Appraisal of Guideline Recommendations on Systemic Therapies for Advanced Hepatocellular Carcinoma: A Review. *JAMA oncology*. [PMID 37535375](#)

Reviews

- **2018** Fukumura D et al. — Enhancing cancer immunotherapy using antiangiogenics: opportunities and challenges. *Nature reviews. Clinical oncology*. [PMID 29508855](#)
- **2019** Jing X et al. — Role of hypoxia in cancer therapy by regulating the tumor microenvironment. *Molecular cancer*. [PMID 31711497](#) FT
- **2023** Chen Z et al. — Hypoxic microenvironment in cancer: molecular mechanisms and therapeutic interventions. *Signal transduction and targeted therapy*. [PMID 36797231](#) FT

Preclinical (in vivo)

- **2010** Rouschop KM et al. — The unfolded protein response protects human tumor cells during hypoxia through regulation of the autophagy genes MAP1LC3B and ATG5. *The Journal of clinical investigation*. [PMID 20038797](#)
- **2014** Noman MZ et al. — PD-L1 is a novel direct target of HIF-1 α , and its blockade under hypoxia enhanced MDSC-mediated T cell activation. *The Journal of experimental medicine*. [PMID 24778419](#) FT

- **2017** Allen E et al. — Combined antiangiogenic and anti-PD-L1 therapy stimulates tumor immunity through HEV formation. Science translational medicine. [PMID 28404866](#)

Mechanistic

- **2015** Voron T et al. — VEGF-A modulates expression of inhibitory checkpoints on CD8+ T cells in tumors. The Journal of experimental medicine. [PMID 25601652](#) FT

Metabolic TME: adenosine, tryptophan, lactate, lipids

13 papers · 2010–2025 · 9 resistance-focused

Clinical trials

- **2019** Long GV et al. — Epcadostat plus pembrolizumab versus placebo plus pembrolizumab in patients with unresectable or metastatic melanoma (ECHO-301/KEYNOTE-252): a phase 3, randomised, double-blind study. The Lancet. Oncology. [PMID 31221619](#)
- **2020** Fong L et al. — Adenosine 2A Receptor Blockade as an Immunotherapy for Treatment-Refractory Renal Cell Cancer. Cancer discovery. [PMID 31732494](#)
- **2024** Tong H et al. — Dual impacts of serine/glycine-free diet in enhancing antitumor immunity and promoting evasion via PD-L1 lactylation. Cell metabolism. [PMID 39577415](#)

Human translational

- **2018** Cascone T et al. — Increased Tumor Glycolysis Characterizes Immune Resistance to Adoptive T Cell Therapy. Cell metabolism. [PMID 29628419](#)

Reviews

- **2021** Bian X et al. — Lipid metabolism and cancer. The Journal of experimental medicine. [PMID 33601415](#) FT
- **2023** Xia C et al. — CD39/CD73/A2AR pathway and cancer immunotherapy. Molecular cancer. [PMID 36859386](#) FT
- **2024** Yan J et al. — Molecular mechanisms and therapeutic significance of Tryptophan Metabolism and signaling in cancer. Molecular cancer. [PMID 39472902](#) FT
- **2025** Chen J et al. — Lactate and lactylation in cancer. Signal transduction and targeted therapy. [PMID 39934144](#) FT

Preclinical (in vivo)

- **2010** Stagg J et al. — Anti-CD73 antibody therapy inhibits breast tumor growth and metastasis. Proceedings of the National Academy of Sciences of the United States of America. [PMID 20080644](#)
- **2016** Yang W et al. — Potentiating the antitumour response of CD8(+) T cells by modulating cholesterol metabolism. Nature. [PMID 26982734](#)
- **2022** Wan W et al. — METTL3/IGF2BP3 axis inhibits tumor immune surveillance by upregulating N6-methyladenosine modification of PD-L1 mRNA in breast cancer. Molecular cancer. [PMID 35197058](#) FT

- **2025** Sun K et al. — Elevated protein lactylation promotes immunosuppressive microenvironment and therapeutic resistance in pancreatic ductal adenocarcinoma. The Journal of clinical investigation. [PMID 39883522](#) FT

Mechanistic

- **2024** Xiao J et al. — 25-Hydroxycholesterol regulates lysosome AMP kinase activation and metabolic reprogramming to educate immunosuppressive macrophages. Immunity. [PMID 38640930](#)

Innate sensing agonists: STING, TLR, RIG-I, type-I IFN

13 papers · 2005–2025 · 4 resistance-focused

Clinical trials

- **2005** Speiser DE et al. — Rapid and strong human CD8+ T cell responses to vaccination with peptide, IFA, and CpG oligodeoxynucleotide 7909. The Journal of clinical investigation. [PMID 15696196](#)
- **2016** Corrales L et al. — The host STING pathway at the interface of cancer and immunity. The Journal of clinical investigation. [PMID 27367184](#)
- **2020** Lv M et al. — Manganese is critical for antitumor immune responses via cGAS-STING and improves the efficacy of clinical immunotherapy. Cell research. [PMID 32839553](#) FT

Human translational

- **2024** Wang J et al. — STING licensing of type I dendritic cells potentiates antitumor immunity. Science immunology. [PMID 38363830](#)

Reviews

- **2020** Kwon J et al. — The Cytosolic DNA-Sensing cGAS-STING Pathway in Cancer. Cancer discovery. [PMID 31852718](#)
- **2020** McLaughlin M et al. — Inflammatory microenvironment remodelling by tumour cells after radiotherapy. Nature reviews. Cancer. [PMID 32161398](#)
- **2021** Amouzegar A et al. — STING Agonists as Cancer Therapeutics. Cancers. [PMID 34070756](#) FT

Preclinical (in vivo)

- **2014** Woo SR et al. — STING-dependent cytosolic DNA sensing mediates innate immune recognition of immunogenic tumors. Immunity. [PMID 25517615](#)
- **2017** Harding SM et al. — Mitotic progression following DNA damage enables pattern recognition within micronuclei. Nature. [PMID 28759889](#) FT
- **2019** Sen T et al. — Targeting DNA Damage Response Promotes Antitumor Immunity through STING-Mediated T-cell Activation in Small Cell Lung Cancer. Cancer discovery. [PMID 30777870](#)

- **2024** Wang Y et al. — Universal STING mimic boosts antitumour immunity via preferential activation of tumour control signalling pathways. *Nature nanotechnology*. [PMID 38480836](#)
- **2025** Lai P et al. — Mitochondrial DNA released by senescent tumor cells enhances PMN-MDSC-driven immunosuppression through the cGAS-STING pathway. *Immunity*. [PMID 40203808](#)

Mechanistic

- **2019** Shae D et al. — Endosomolytic polymersomes increase the activity of cyclic dinucleotide STING agonists to enhance cancer immunotherapy. *Nature nanotechnology*. [PMID 30664751](#) FT

Engineered & localised cytokines

13 papers · 2012–2025 · 5 resistance-focused

Clinical trials

- **2017** Ridker PM et al. — Effect of interleukin-1 β inhibition with canakinumab on incident lung cancer in patients with atherosclerosis: exploratory results from a randomised, double-blind, placebo-controlled trial. *Lancet (London, England)*. [PMID 28855077](#)
- **2018** Wrangle JM et al. — ALT-803, an IL-15 superagonist, in combination with nivolumab in patients with metastatic non-small cell lung cancer: a non-randomised, open-label, phase 1b trial. *The Lancet. Oncology*. [PMID 29628312](#)
- **2019** Chiocca EA et al. — Regulatable interleukin-12 gene therapy in patients with recurrent high-grade glioma: Results of a phase 1 trial. *Science translational medicine*. [PMID 31413142](#)

Reviews

- **2012** Hao NB et al. — Macrophages in tumor microenvironments and the progression of tumors. *Clinical & developmental immunology*. [PMID 22778768](#) FT
- **2015** Tugues S et al. — New insights into IL-12-mediated tumor suppression. *Cell death and differentiation*. [PMID 25190142](#)
- **2019** Conlon KC et al. — Cytokines in the Treatment of Cancer. *Journal of interferon & cytokine research : the official journal of the International Society for Interferon and Cytokine Research*. [PMID 29889594](#)

Preclinical (in vivo)

- **2020** Weiss T et al. — Immunocytokines are a promising immunotherapeutic approach against glioblastoma. *Science translational medicine*. [PMID 33028706](#)
- **2021** Agliardi G et al. — Intratumoral IL-12 delivery empowers CAR-T cell immunotherapy in a pre-clinical model of glioblastoma. *Nature communications*. [PMID 33469002](#) FT
- **2023** Bhatia V et al. — Targeting advanced prostate cancer with STEAP1 chimeric antigen receptor T cell and tumor-localized IL-12 immunotherapy. *Nature communications*. [PMID 37041154](#) FT

- **2024** Landoni E et al. — IL-12 reprograms CAR-expressing natural killer T cells to long-lived Th1-polarized cells with potent antitumor activity. Nature communications. PMID 38167707 FT
- **2024** Shiri AM et al. — IL-10 dampens antitumor immunity and promotes liver metastasis via PD-L1 induction. Journal of hepatology. PMID 38160941 FT
- **2025** Zotta A et al. — Mitochondrial respiratory complex III sustains IL-10 production in activated macrophages and promotes tumor-mediated immune evasion. Science advances. PMID 39841829 FT

Mechanistic

- **2014** Ruffell B et al. — Macrophage IL-10 blocks CD8+ T cell-dependent responses to chemotherapy by suppressing IL-12 expression in intratumoral dendritic cells. Cancer cell. PMID 25446896

T-cell trafficking, chemokines & tertiary lymphoid structures

13 papers · 2005–2025 · 4 resistance-focused

Clinical trials

- **2012** Liu S et al. — CD8+ lymphocyte infiltration is an independent favorable prognostic indicator in basal-like breast cancer. Breast cancer research : BCR. PMID 22420471 FT
- **2020** Bockorny B et al. — BL-8040, a CXCR4 antagonist, in combination with pembrolizumab and chemotherapy for pancreatic cancer: the COMBAT trial. Nature medicine. PMID 32451495
- **2025** Osorio JC et al. — Fc-optimized CD40 agonistic antibody elicits tertiary lymphoid structure formation and systemic antitumor immunity in metastatic cancer. Cancer cell. PMID 40816292

Human translational

- **2020** Helmink BA et al. — B cells and tertiary lymphoid structures promote immunotherapy response. Nature. PMID 31942075

Reviews

- **2006** Burger JA et al. — CXCR4: a key receptor in the crosstalk between tumor cells and their microenvironment. Blood. PMID 16269611
- **2015** Teng MW et al. — Classifying Cancers Based on T-cell Infiltration and PD-L1. Cancer research. PMID 25977340
- **2024** Yi M et al. — Targeting cytokine and chemokine signaling pathways for cancer therapy. Signal transduction and targeted therapy. PMID 39034318 FT
- **2024** Mempel TR et al. — How chemokines organize the tumour microenvironment. Nature reviews. Cancer. PMID 38066335

Preclinical (in vivo)

- **2005** Orimo A et al. — Stromal fibroblasts present in invasive human breast carcinomas promote tumor growth and angiogenesis through elevated SDF-1/CXCL12 secretion. *Cell*. PMID 15882617
- **2013** Feig C et al. — Targeting CXCL12 from FAP-expressing carcinoma-associated fibroblasts synergizes with anti-PD-L1 immunotherapy in pancreatic cancer. *Proceedings of the National Academy of Sciences of the United States of America*. PMID 24277834
- **2013** Gu-Trantien C et al. — CD4⁺ follicular helper T cell infiltration predicts breast cancer survival. *The Journal of clinical investigation*. PMID 23778140
- **2024** Lim RJ et al. — CXCL9/10-engineered dendritic cells promote T cell activation and enhance immune checkpoint blockade for lung cancer. *Cell reports. Medicine*. PMID 38518770 FT
- **2025** Amisaki M et al. — IL-33-activated ILC2s induce tertiary lymphoid structures in pancreatic cancer. *Nature*. PMID 39814891 FT

Antigen presentation, MHC-I loss & IFN-γ pathway defects

13 papers · 2006–2025 · 6 resistance-focused

Clinical trials

- **2017** Ott PA et al. — An immunogenic personal neoantigen vaccine for patients with melanoma. *Nature*. PMID 28678778
- **2018** Skoulidis F et al. — STK11/LKB1 Mutations and PD-1 Inhibitor Resistance in KRAS-Mutant Lung Adenocarcinoma. *Cancer discovery*. PMID 29773717
- **2023** Rojas LA et al. — Personalized RNA neoantigen vaccines stimulate T cells in pancreatic cancer. *Nature*. PMID 37165196 FT
- **2024** Weber JS et al. — Individualised neoantigen therapy mRNA-4157 (V940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (KEYNOTE-942): a randomised, phase 2b study. *Lancet (London, England)*. PMID 38246194
- **2025** Sethna Z et al. — RNA neoantigen vaccines prime long-lived CD8⁺ T cells in pancreatic cancer. *Nature*. PMID 39972124 FT
- **2025** Braun DA et al. — A neoantigen vaccine generates antitumour immunity in renal cell carcinoma. *Nature*. PMID 39910301 FT

Human translational

- **2017** Shin DS et al. — Primary Resistance to PD-1 Blockade Mediated by JAK1/2 Mutations. *Cancer discovery*. PMID 27903500

Reviews

- **2021** Blass E et al. — Advances in the development of personalized neoantigen-based therapeutic cancer vaccines. *Nature reviews. Clinical oncology*. PMID 33473220 FT
- **2023** Xie N et al. — Neoantigens: promising targets for cancer therapy. *Signal transduction and targeted therapy*. PMID 36604431 FT

- **2023** Wu X et al. — Targeting MHC-I molecules for cancer: function, mechanism, and therapeutic prospects. *Molecular cancer*. [PMID 38041084](#) FT

Preclinical (in vivo)

- **2006** Reits EA et al. — Radiation modulates the peptide repertoire, enhances MHC class I expression, and induces successful antitumor immunotherapy. *The Journal of experimental medicine*. [PMID 16636135](#) FT
- **2016** Koyama S et al. — STK11/LKB1 Deficiency Promotes Neutrophil Recruitment and Proinflammatory Cytokine Production to Suppress T-cell Activity in the Lung Tumor Microenvironment. *Cancer research*. [PMID 26833127](#)
- **2020** Yamamoto K et al. — Autophagy promotes immune evasion of pancreatic cancer by degrading MHC-I. *Nature*. [PMID 32376951](#) FT

Epigenetic priming & viral mimicry

13 papers · 2014-2025 · 8 resistance-focused

Clinical trials

- **2014** Odunsi K et al. — Epigenetic potentiation of NY-ESO-1 vaccine therapy in human ovarian cancer. *Cancer immunology research*. [PMID 24535937](#)
- **2021** Ny L et al. — The PEMDAC phase 2 study of pembrolizumab and entinostat in patients with metastatic uveal melanoma. *Nature communications*. [PMID 34453044](#) FT
- **2024** Baretta M et al. — Entinostat in combination with nivolumab in metastatic pancreatic ductal adenocarcinoma: a phase 2 clinical trial. *Nature communications*. [PMID 39532835](#) FT

Reviews

- **2019** Emran AA et al. — Targeting DNA Methylation and EZH2 Activity to Overcome Melanoma Resistance to Immunotherapy. *Trends in immunology*. [PMID 30853334](#)
- **2021** Wong KK — DNMT1: A key drug target in triple-negative breast cancer. *Seminars in cancer biology*. [PMID 32461152](#)
- **2024** Zhou L et al. — Epigenetic modulations in triple-negative breast cancer: Therapeutic implications for tumor microenvironment. *Pharmacological research*. [PMID 38719195](#)

Preclinical (in vivo)

- **2015** Woods DM et al. — HDAC Inhibition Upregulates PD-1 Ligands in Melanoma and Augments Immunotherapy with PD-1 Blockade. *Cancer immunology research*. [PMID 26297712](#)
- **2018** Sheng W et al. — LSD1 Ablation Stimulates Anti-tumor Immunity and Enables Checkpoint Blockade. *Cell*. [PMID 29937226](#)
- **2021** Zhang SM et al. — KDM5B promotes immune evasion by recruiting SETDB1 to silence retroelements. *Nature*. [PMID 34671158](#)

Mechanistic

- **2015** Chiappinelli KB et al. — Inhibiting DNA Methylation Causes an Interferon Response in Cancer via dsRNA Including Endogenous Retroviruses. Cell. PMID 26317466
- **2025** Porazzi P et al. — EZH1/EZH2 inhibition enhances adoptive T cell immunotherapy against multiple cancer models. Cancer cell. PMID 39983725
- **2025** Jiang Q et al. — HIF regulates multiple translated endogenous retroviruses: Implications for cancer immunotherapy. Cell. PMID 40023154 FT
- **2025** Aine M et al. — The DNA methylation landscape of primary triple-negative breast cancer. Nature communications. PMID 40155623 FT

Radiation & chemotherapy priming, immunogenic cell death

13 papers · 2009–2025 · 6 resistance-focused

Clinical trials

- **2014** Kwon ED et al. — Ipilimumab versus placebo after radiotherapy in patients with metastatic castration-resistant prostate cancer that had progressed after docetaxel chemotherapy (CA184-043): a multicentre, randomised, double-blind, phase 3 trial. The Lancet. Oncology. PMID 24831977
- **2018** Formenti SC et al. — Radiotherapy induces responses of lung cancer to CTLA-4 blockade. Nature medicine. PMID 30397353
- **2019** Theelen WSME et al. — Effect of Pembrolizumab After Stereotactic Body Radiotherapy vs Pembrolizumab Alone on Tumor Response in Patients With Advanced Non-Small Cell Lung Cancer: Results of the PEMBRO-RT Phase 2 Randomized Clinical Trial. JAMA oncology. PMID 31294749
- **2022** Lim M et al. — Phase III trial of chemoradiotherapy with temozolomide plus nivolumab or placebo for newly diagnosed glioblastoma with methylated MGMT promoter. Neuro-oncology. PMID 35511454 FT
- **2025** Huang J et al. — Combination of pembrolizumab and radiotherapy induces systemic antitumor immune responses in immunologically cold non-small cell lung cancer. Nature cancer. PMID 40696153 FT

Human translational

- **2024** Shiao SL et al. — Single-cell and spatial profiling identify three response trajectories to pembrolizumab and radiation therapy in triple negative breast cancer. Cancer cell. PMID 38194915

Reviews

- **2015** Barker HE et al. — The tumour microenvironment after radiotherapy: mechanisms of resistance and recurrence. Nature reviews. Cancer. PMID 26105538
- **2019** Duan X et al. — Nanoparticle-Mediated Immunogenic Cell Death Enables and Potentiates Cancer Immunotherapy. Angewandte Chemie (International ed. in English). PMID 30016571

- **2022** Zhang Z et al. — Radiotherapy combined with immunotherapy: the dawn of cancer treatment. Signal transduction and targeted therapy. [PMID 35906199](#) FT
- **2024** Meier P et al. — Immunogenic cell death in cancer: targeting necroptosis to induce antitumour immunity. Nature reviews. Cancer. [PMID 38454135](#)

Preclinical (in vivo)

- **2009** Dewan MZ et al. — Fractionated but not single-dose radiotherapy induces an immune-mediated abscopal effect when combined with anti-CTLA-4 antibody. Clinical cancer research : an official journal of the American Association for Cancer Research. [PMID 19706802](#)
- **2018** Kuai R et al. — Elimination of established tumors with nanodisc-based combination chemoimmunotherapy. Science advances. [PMID 29675465](#) FT
- **2019** Turubanova VD et al. — Immunogenic cell death induced by a new photodynamic therapy based on photosens and photodithazine. Journal for immunotherapy of cancer. [PMID 31842994](#) FT

Intratumoral therapy, oncolytic viruses & engineered bacteria

13 papers · 2010-2025 · 3 resistance-focused

Clinical trials

- **2017** Ribas A et al. — Oncolytic Virotherapy Promotes Intratumoral T Cell Infiltration and Improves Anti-PD-1 Immunotherapy. Cell. [PMID 28886381](#)
- **2019** Cloughesy TF et al. — Neoadjuvant anti-PD-1 immunotherapy promotes a survival benefit with intratumoral and systemic immune responses in recurrent glioblastoma. Nature medicine. [PMID 30742122](#) FT
- **2021** Friedman GK et al. — Oncolytic HSV-1 G207 Immunovirotherapy for Pediatric High-Grade Gliomas. The New England journal of medicine. [PMID 33838625](#)
- **2023** Ling AL et al. — Clinical trial links oncolytic immunoactivation to survival in glioblastoma. Nature. [PMID 37853118](#) FT
- **2024** Galanis E et al. — Carcinoembryonic antigen-expressing oncolytic measles virus derivative in recurrent glioblastoma: a phase 1 trial. Nature communications. [PMID 38216554](#) FT
- **2025** Shen Y et al. — Oncolytic virus VG161 in refractory hepatocellular carcinoma. Nature. [PMID 40108464](#)

Reviews

- **2017** Lawler SE et al. — Oncolytic Viruses in Cancer Treatment: A Review. JAMA oncology. [PMID 27441411](#)
- **2018** Bommarreddy PK et al. — Integrating oncolytic viruses in combination cancer immunotherapy. Nature reviews. Immunology. [PMID 29743717](#)
- **2020** Macedo N et al. — Clinical landscape of oncolytic virus research in 2020. Journal for immunotherapy of cancer. [PMID 33046622](#) FT

Preclinical (in vivo)

- **2010** Cerullo V et al. — Oncolytic adenovirus coding for granulocyte macrophage colony-stimulating factor induces antitumoral immunity in cancer patients. *Cancer research*. PMID 20484030
- **2016** Scharping NE et al. — The Tumor Microenvironment Represses T Cell Mitochondrial Biogenesis to Drive Intratumoral T Cell Metabolic Insufficiency and Dysfunction. *Immunity*. PMID 27496732
- **2017** Zheng JH et al. — Two-step enhanced cancer immunotherapy with engineered *Salmonella typhimurium* secreting heterologous flagellin. *Science translational medicine*. PMID 28179508

Mechanistic

- **2024** Espinosa-Carrasco G et al. — Intratumoral immune triads are required for immunotherapy-mediated elimination of solid tumors. *Cancer cell*. PMID 38906155

Microbiome modulation & FMT

13 papers · 2015–2026 · 6 resistance-focused

Clinical trials

- **2021** Davar D et al. — Fecal microbiota transplant overcomes resistance to anti-PD-1 therapy in melanoma patients. *Science (New York, N.Y.)*. PMID 33542131
- **2021** Baruch EN et al. — Fecal microbiota transplant promotes response in immunotherapy-refractory melanoma patients. *Science (New York, N.Y.)*. PMID 33303685
- **2023** Routy B et al. — Fecal microbiota transplantation plus anti-PD-1 immunotherapy in advanced melanoma: a phase I trial. *Nature medicine*. PMID 37414899
- **2026** Duttagupta S et al. — Fecal microbiota transplantation plus immunotherapy in non-small cell lung cancer and melanoma: the phase 2 FMT-LUMINate trial. *Nature medicine*. PMID 41606121 FT

Human translational

- **2019** Riquelme E et al. — Tumor Microbiome Diversity and Composition Influence Pancreatic Cancer Outcomes. *Cell*. PMID 31398337
- **2024** Derosa L et al. — Custom scoring based on ecological topology of gut microbiota associated with cancer immunotherapy outcome. *Cell*. PMID 38906102

Reviews

- **2023** Wong CC et al. — Gut microbiota in colorectal cancer development and therapy. *Nature reviews. Clinical oncology*. PMID 37169888
- **2024** Routy B et al. — Melanoma and microbiota: Current understanding and future directions. *Cancer cell*. PMID 38157864
- **2024** Yadegar A et al. — Fecal microbiota transplantation: current challenges and future landscapes. *Clinical microbiology reviews*. PMID 38717124

Preclinical (in vivo)

- **2015** Vétizou M et al. — Anticancer immunotherapy by CTLA-4 blockade relies on the gut microbiota. Science (New York, N.Y.). [PMID 26541610](#)
- **2018** Pushalkar S et al. — The Pancreatic Cancer Microbiome Promotes Oncogenesis by Induction of Innate and Adaptive Immune Suppression. Cancer discovery. [PMID 29567829](#)
- **2022** Huang J et al. — Ginseng polysaccharides alter the gut microbiota and kynurenine/tryptophan ratio, potentiating the antitumor effect of anti-programmed cell death 1/programmed cell death ligand 1 (anti-PD-1/PD-L1) immunotherapy. Gut. [PMID 34006584](#) FT
- **2024** Wang M et al. — Killing tumor-associated bacteria with a liposomal antibiotic generates neoantigens that induce anti-tumor immune responses. Nature biotechnology. [PMID 37749267](#)

TME-directed delivery & regulated cell death

13 papers · 2017–2025 · 6 resistance-focused

Clinical trials

- **2024** Xia Y et al. — PARP inhibitors enhance antitumor immune responses by triggering pyroptosis via TNF-caspase 8-GSDMD/E axis in ovarian cancer. Journal for immunotherapy of cancer. [PMID 39366751](#) FT

Human translational

- **2024** Cheng X et al. — Atlas of Metastatic Gastric Cancer Links Ferroptosis to Disease Progression and Immunotherapy Response. Gastroenterology. [PMID 39097198](#)

Reviews

- **2022** Tong X et al. — Targeting cell death pathways for cancer therapy: recent developments in necroptosis, pyroptosis, ferroptosis, and cuproptosis research. Journal of hematology & oncology. [PMID 36482419](#) FT
- **2024** Lei G et al. — The roles of ferroptosis in cancer: Tumor suppression, tumor microenvironment, and therapeutic interventions. Cancer cell. [PMID 38593779](#)
- **2024** Wang H et al. — Crosstalk of pyroptosis and cytokine in the tumor microenvironment: from mechanisms to clinical implication. Molecular cancer. [PMID 39614288](#) FT
- **2025** Sabit H et al. — Precision nanomedicine: navigating the tumor microenvironment for enhanced cancer immunotherapy and targeted drug delivery. Molecular cancer. [PMID 40457437](#) FT

Preclinical (in vivo)

- **2017** Yang G et al. — Hollow MnO₂ as a tumor-microenvironment-responsive biodegradable nano-platform for combination therapy favoring antitumor immune responses. Nature communications. [PMID 29026068](#) FT

- **2019** Wang W et al. — CD8+ T cells regulate tumour ferroptosis during cancer immunotherapy. *Nature*. [PMID 31043744](#)
- **2022** Kim R et al. — Ferroptosis of tumour neutrophils causes immune suppression in cancer. *Nature*. [PMID 36385526](#)
- **2023** Li J et al. — Tumor-specific GPX4 degradation enhances ferroptosis-initiated antitumor immune response in mouse models of pancreatic cancer. *Science translational medicine*. [PMID 37910602](#)
- **2024** Ma L et al. — Targeting carnitine palmitoyl transferase 1A (CPT1A) induces ferroptosis and synergizes with immunotherapy in lung cancer. *Signal transduction and targeted therapy*. [PMID 38453925](#)
- **2024** Zhou Y et al. — Mitochondria-targeted photodynamic therapy triggers GSDME-mediated pyroptosis and sensitizes anti-PD-1 therapy in colorectal cancer. *Journal for immunotherapy of cancer*. [PMID 38429070](#) FT

Mechanistic

- **2023** Yang F et al. — Ferroptosis heterogeneity in triple-negative breast cancer reveals an innovative immunotherapy combination strategy. *Cell metabolism*. [PMID 36257316](#)

Cell therapy against the solid-tumour TME

13 papers · 2015–2025 · 6 resistance-focused

Clinical trials

- **2015** Ahmed N et al. — Human Epidermal Growth Factor Receptor 2 (HER2) -Specific Chimeric Antigen Receptor-Modified T Cells for the Immunotherapy of HER2-Positive Sarcoma. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*. [PMID 25800760](#)
- **2017** O'Rourke DM et al. — A single dose of peripherally infused EGFRvIII-directed CAR T cells mediates antigen loss and induces adaptive resistance in patients with recurrent glioblastoma. *Science translational medicine*. [PMID 28724573](#)
- **2022** Narayan V et al. — PSMA-targeting TGFβ-insensitive armored CAR T cells in metastatic castration-resistant prostate cancer: a phase 1 trial. *Nature medicine*. [PMID 35314843](#)
- **2024** Schoenfeld AJ et al. — Lifleucel, an Autologous Tumor-Infiltrating Lymphocyte Monotherapy, in Patients with Advanced Non-Small Cell Lung Cancer Resistant to Immune Checkpoint Inhibitors. *Cancer discovery*. [PMID 38563600](#) FT
- **2024** Bagley SJ et al. — Repeated peripheral infusions of anti-EGFRvIII CAR T cells in combination with pembrolizumab show no efficacy in glioblastoma: a phase 1 trial. *Nature cancer*. [PMID 38216766](#)

Reviews

- **2021** Sterner RC et al. — CAR-T cell therapy: current limitations and potential strategies. *Blood cancer journal*. [PMID 33824268](#) FT
- **2023** Maalej KM et al. — CAR-cell therapy in the era of solid tumor treatment: current challenges and emerging therapeutic advances. *Molecular cancer*. [PMID 36717905](#) FT

- **2025** Zugasti I et al. — CAR-T cell therapy for cancer: current challenges and future directions. Signal transduction and targeted therapy. [PMID 40610404](#) FT

Preclinical (in vivo)

- **2018** Kagoya Y et al. — A novel chimeric antigen receptor containing a JAK-STAT signaling domain mediates superior antitumor effects. Nature medicine. [PMID 29400710](#)
- **2020** Klichinsky M et al. — Human chimeric antigen receptor macrophages for cancer immunotherapy. Nature biotechnology. [PMID 32361713](#)
- **2020** Amor C et al. — Senolytic CAR T cells reverse senescence-associated pathologies. Nature. [PMID 32555459](#) FT
- **2024** Zhao Y et al. — IL-10-expressing CAR T cells resist dysfunction and mediate durable clearance of solid tumors and metastases. Nature biotechnology. [PMID 38168996](#)
- **2024** Chan JD et al. — FOXO1 enhances CAR T cell stemness, metabolic fitness and efficacy. Nature. [PMID 38600376](#) FT

Physical & energy-based TME modulation

13 papers · 2002–2024 · 8 resistance-focused

Clinical trials

- **2002** Wust P et al. — Hyperthermia in combined treatment of cancer. The Lancet. Oncology. [PMID 12147435](#)
- **2019** Kobayashi H et al. — Near-Infrared Photoimmunotherapy of Cancer. Accounts of chemical research. [PMID 31335117](#) FT
- **2021** Chen KT et al. — Neuronavigation-guided focused ultrasound for transcranial blood-brain barrier opening and immunostimulation in brain tumors. Science advances. [PMID 33547073](#) FT
- **2024** Datta P et al. — Recent Advances and Future Directions in Sonodynamic Therapy for Cancer Treatment. BME frontiers. [PMID 39735354](#) FT

Reviews

- **2020** Son S et al. — Multifunctional sonosensitizers in sonodynamic cancer therapy. Chemical Society reviews. [PMID 32337527](#)
- **2021** Alzeibak R et al. — Targeting immunogenic cancer cell death by photodynamic therapy: past, present and future. Journal for immunotherapy of cancer. [PMID 33431631](#) FT
- **2023** Overchuk M et al. — Photodynamic and Photothermal Therapies: Synergy Opportunities for Nanomedicine. ACS nano. [PMID 37129253](#) FT
- **2024** Obrador E et al. — Glioblastoma Therapy: Past, Present and Future. International journal of molecular sciences. [PMID 38473776](#) FT

Preclinical (in vivo)

- **2012** Waitz R et al. — Potent induction of tumor immunity by combining tumor cryoablation with anti-CTLA-4 therapy. Cancer research. [PMID 22108823](#)
- **2023** Li Y et al. — Nanosensitizer-mediated augmentation of sonodynamic therapy efficacy and antitumor immunity. Nature communications. [PMID 37914681](#) FT
- **2024** Wang Y et al. — Tumor Cell-Targeting and Tumor Microenvironment-Responsive Nanoplatforms for the Multimodal Imaging-Guided Photodynamic/Photothermal/Chemodynamic Treatment of Cervical Cancer. International journal of nanomedicine. [PMID 38887692](#) FT

Mechanistic

- **2016** He C et al. — Core-shell nanoscale coordination polymers combine chemotherapy and photodynamic therapy to potentiate checkpoint blockade cancer immunotherapy. Nature communications. [PMID 27530650](#) FT
- **2024** Li B et al. — γ -Glutamyl transpeptidase-activable nanoprobe crosses the blood-brain barrier for immuno-sonodynamic therapy of glioma. Nature communications. [PMID 39613729](#) FT

Spatial and single-cell dissection of the TME

13 papers · 2019–2025 · 4 resistance-focused

Clinical trials

- **2019** Wagner J et al. — A Single-Cell Atlas of the Tumor and Immune Ecosystem of Human Breast Cancer. Cell. [PMID 30982598](#) FT
- **2023** Song X et al. — Spatial multi-omics revealed the impact of tumor ecosystem heterogeneity on immunotherapy efficacy in patients with advanced non-small cell lung cancer treated with bispecific antibody. Journal for immunotherapy of cancer. [PMID 36854570](#) FT
- **2023** Zhang G et al. — Multi-omics analysis uncovers tumor ecosystem dynamics during neoadjuvant toripalimab plus nab-paclitaxel and S-1 for esophageal squamous cell carcinoma: a single-center, open-label, single-arm phase 2 trial. EBioMedicine. [PMID 36921563](#) FT

Human translational

- **2022** Sammut SJ et al. — Multi-omic machine learning predictor of breast cancer therapy response. Nature. [PMID 34875674](#) FT
- **2024** Du Y et al. — Integration of Pan-Cancer Single-Cell and Spatial Transcriptomics Reveals Stromal Cell Features and Therapeutic Targets in Tumor Microenvironment. Cancer research. [PMID 38225927](#)
- **2025** Liu Z et al. — A single-cell atlas reveals immune heterogeneity in anti-PD-1-treated non-small cell lung cancer. Cell. [PMID 40147443](#)

Reviews

- **2020** Ding S et al. — Single-cell RNA sequencing in breast cancer: Understanding tumor heterogeneity and paving roads to individualized therapy. Cancer communications (London, England). PMID 32654419 FT
- **2024** Feng DC et al. — The implications of single-cell RNA-seq analysis in prostate cancer: unraveling tumor heterogeneity, therapeutic implications and pathways towards personalized therapy. Military Medical Research. PMID 38605399 FT
- **2025** Fu Y et al. — Single-cell and spatial transcriptomic insights into glioma cellular heterogeneity and metabolic adaptations. Frontiers in immunology. PMID 40255400 FT

Preclinical (in vivo)

- **2022** Yeo AT et al. — Single-cell RNA sequencing reveals evolution of immune landscape during glioblastoma progression. Nature immunology. PMID 35624211 FT
- **2023** Jain S et al. — Single-cell RNA sequencing and spatial transcriptomics reveal cancer-associated fibroblasts in glioblastoma with protumoral effects. The Journal of clinical investigation. PMID 36856115 FT
- **2025** Zhu B et al. — Spatial and multiomics analysis of human and mouse lung adenocarcinoma precursors reveals TIM-3 as a putative target for precancer interception. Cancer cell. PMID 40345189

Mechanistic

- **2025** Liu Y et al. — Conserved spatial subtypes and cellular neighborhoods of cancer-associated fibroblasts revealed by single-cell spatial multi-omics. Cancer cell. PMID 40154487